Amal Mehta

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Research engineer with six years of applied deep learning, from generative models and vision to deep reinforcement learning, taking work from data and training to policies running on real robots. Earlier neuroimaging research at UCSF, and a long-standing interest in how the brain learns.

Experience

Independent ML & Robotics Research **Sept 2025 – Present**
Self-directed projects; details at amalmehta.github.io/PortfolioWebsite *San Francisco Bay Area*

- **Reward hacking in code agents:** GRPO on visible tests taught an agent to delete 98% of untested branches (true correctness 31% → 2% as pass rate rose); random 25% hidden-check audits prevented it. Pre-registered and ran 50 follow-up experiments.
- **Brain–computer interfaces:** four projects on public neural data, three with papers; cursor decoding held R^2 0.72 for 138 days.
- **Post-training:** SFT then GRPO on a small transformer policy in a key-and-door gridworld maze; over 3 seeds, GRPO lifted held-out success from 70% to 80% (DPO: 81%), where RL from scratch failed.
- **Meta-RL and self-play:** quadrotor adapting to payload, wind and motor wear with seven meta-learning methods; 9×9 Go by self-play.
- **Tools:** Researcher Pal (Claude Code skills that teach any paper); one PyTorch loop split across DDP, FSDP and pipeline.
- Member, Sensorimotor AI journal club, Redwood Center (UC Berkeley); mentored undergraduate scholars from Berkeley.

Deep Learning Scientist **Sept 2021 – Aug 2025**
PhysicsAI (DARPA-funded) *Pacifica, CA (remote)*

- Formulated synthetic-data generation as RL: policies (TD3, PPO, DDPG) set image-generator parameters, rewarded by detector performance on real drone imagery, closing the sim-to-real gap without hand-tuned rendering.
- Built a PEARL meta-RL agent (probabilistic task embeddings over TD3) for fast few-shot adaptation across generator tasks.
- Implemented domain adaptation from scratch to hold detector performance in the low-real-label, high-synthetic-data regime.
- Built GAN-based synthesis pipelines (DCGAN, InfoGAN, FineGAN) producing training data for downstream detection models.

Deep RL Researcher **Aug 2023 – Mar 2024**
Hybrid Robotics Lab, UC Berkeley — adv. Prof. Koushil Sreenath *Berkeley, CA*

- Trained a single multi-stage PPO policy in IsaacGym for a Unitree Go1 quadruped to track, approach, and shoot a soccer ball at a target, replacing the multi-policy hierarchical framework used in prior work; ∼60% kicking accuracy on hardware.
- Designed the three-stage reward and transition machinery and CPG-based gait tracking; prototyped RLPD (SAC with prior data) to fine-tune the sim-trained policy on real robot data.

Research Assistant **June 2020 – Dec 2022**
Video & Image Processing Lab, BAIR — adv. Prof. Avideh Zakhor *UC Berkeley*

- Developed a deep CNN segmentation pipeline for invasive melanoma in large-scale histopathological images (F1 0.44 vs. 0.14 for the prior best); published at SPIE Medical Imaging 2023; Stowell-Orbison Award Poster Session at USCAP 2022.

Researcher **June 2020 – Aug 2020**
Abbasi-Asl Lab, UCSF — adv. Prof. Reza Abbasi-Asl *San Francisco, CA*

- Built empirical mode decomposition (EMD) pipelines for 2D and 3D brain image registration.

Education

University of California, Berkeley **Aug 2017 – Aug 2021**
B.A. Computer Science — Regents Scholar *Berkeley, CA*

- Focused on robotics and linear system theory; Regents, merit, and leadership awards.
- CS285 (Deep RL) project: iterative retraining of PPO agents against adversarial policies.

Stanford University **Aug 2021 – May 2023**
Graduate coursework (non-degree) — Autonomous Systems & Artificial Intelligence

- CS231N (Deep Learning for Vision), CS330 (Meta-Learning), AA271A/B (Autonomy I & II), AA273 (State Estimation).
- CS330 project: few-shot imitation learning; implemented RL^2 and inverse soft Q-learning in Meta-World.

Technical Skills

Languages	Python, C++, Swift, JavaScript / TypeScript
ML / RL	PyTorch, TensorFlow, stable-baselines3, IsaacGym, OpenCV, CTC decoding
Product	SwiftUI, Core ML, Electron, Three.js, Next.js, React, SQLite
Robotics	ROS, Gazebo, Unitree Go1, CPG gait control, Kalman filtering, Shared autonomy
Tools	Docker, GitHub Actions CI, CUDA/multi-GPU, DDP/FSDP, W&B, Testing (XCTest, Vitest, Playwright)

Publications

1. Shah, A., **Mehta, A.**, Wang, M., Neumann, N., McCalmont, T., Zakhor, A. "Deep Learning Segmentation for Invasive Melanoma." *SPIE Medical Imaging*, 2023.
2. Neumann, N., Wang, M., **Mehta, A.**, et al. "Quantifying Invasive Melanoma Volume by Deep Learning Segmentation at the Pixel-Level." *USCAP Annual Meeting*, 2022.